

- Medical Image Analysis
- Computer Vision
- Representation learning
- AI for healthcare

SUMMARY

Algorithm Engineer at Noah Medical with a Ph.D. in Biomedical Engineering from Rensselaer Polytechnic Institute. Focused on developing deep learning algorithms for medical image analysis and computer vision, with experience spanning research innovation and product delivery.

EDUCATION

Ph.D., Department of Biomedical Engineering, Rensselaer Polytechnic Institute 08/2018 - 12/2023
B.S., School of Computing, Wuhan University 09/2014 - 06/2018

CAMPUS EXPERIENCE

Research Assistant, Deep Imaging Analytics Lab, Rensselaer Polytechnic Institute, Troy 09/2018 - 11/2023

- Advised by Dr. Pingkun Yan
- Identified clinically important problems with doctors, and developed possible DL-based solutions in collaboration with clinical sites, i.e. National Institutes of Health (NIH), Massachusetts General Hospital (MGH), Houston Methodist Research Institute (HMRI).

Research Intern, State Key Laboratory of Intelligent Technology and Systems, Tsinghua University, Beijing 09/2017 - 02/2018

- Advised by Dr. Xiaolin Hu
- Completed image captioning task by developing a new attention method based on saliency.

Research Team Member, The School of Computing, Wuhan University, Wuhan 09/2016 - 08/2017

- Advised by Dr. Bo Du
- Applied popular deep neural networks, including FCN, GAN and U-Net to segment the prostate in magnetic resonance imaging (MRI).

INDUSTRY EXPERIENCE

Algorithm Engineer Imaging and Machine Learning Team, Noah Medical, San Carlos 12/2023 - Present

- Worked with Dr. Tao Zhao and Dr. Haili Chui on improving the quality of imaging in the Galaxy system.
- Developed a fluoro image sequence annotation tool using the AI tracking algorithm, which reduced annotation time by more than 80%.
- Built a traceable ML workflow with experiment tracking and model versioning using MLflow and Weights & Biases, improving reproducibility and model comparison across iterations.
- Led and managed the end-to-end development lifecycle for project features: requirement definition, technical specification, software design document, unit testing, verification, and clinical validation.
- Led projects and managed the entire deep learning solution lifecycle—from developing, training, and evaluating models to integrating, deploying, and testing them for end-to-end product delivery.

Immersive Media Research Intern Immersive Media Lab, Tencent America, Palo Alto 02/2022 - 05/2022

- Worked with Dr. Bing Jian and Dr. Lu He on 3D Virtual Tour.
- Completed a pipeline for 360° image panoptic segmentation through unsupervised domain adaptation.

Deep Learning Research Intern Infervision, Beijing 03/2017 - 05/2017

- Worked on a CT nodule detection project cooperated with Tongji Hospital, Wuhan.
- Applied and modified Faster R-CNN to detect lung nodules in MRI and chest X-ray images.

PUBLICATIONS

Conference

- **Patient-specific reference model estimation for orthognathic surgical planning.** Fang X et al. In: 2024 International Conference on Information Processing in Computer-Assisted Interventions (IPCAI 2024) (Early Acceptance, IJCARS special issue, **The NVIDIA Bench-to-Bedside Award Runner-up**)
- **PGTA: Prior-Guided Tri-Axial Attentive Pretraining for Esophageal Cancer Segmentation in Small-Sample Settings.** Shi Q, Duan W, Zhang A, Kang L, Ke Q, Yang H, Lu H, Yuan J, Li H, Fang X**, Wang R**, Ruan L**, Wan X** (**Corresponding author). In: *IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR 2024)*, Under review
- **Soft-tissue Driven Craniomaxillofacial Surgical Planning.** Fang X et al. In: 2023 International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI 2023) (Early Acceptance, **Top 13.6%**)

- **Spatiotemporal Incremental Mechanics Modeling of Facial Tissue Change.** Lampen N, Kim D, Xu X, Fang X et al. In: 2023 *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI 2023)* (Early Acceptance, Top 13.6%)
- **TriAxial Low-Rank Transformer for Efficient Medical Image Segmentation.** Shang J, and Fang X** (**Corresponding author). In: 2023 *Chinese Conference on Pattern Recognition and Computer Vision (PRCV 2023)*
- **Deep Learning-Based Facial Appearance Simulation Driven by Surgically Planned Craniomaxillofacial Bony Movement.** Fang X et al. In: 2022 *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI 2022)* (Oral presentation, MICCAI Young Scientist Award (Top 0.3%), MICCAI Travel Award, Best of MICCAI)
- **Division and Fusion: Rethink Convolutional Kernels for 3D Medical Image Segmentation.** Fang X et al. In: 2020 *Machine Learning in Medical Imaging International Workshop (MICCAI MLMI 2020)* (Oral presentation)
- **Unified multi-scale feature abstraction for medical image segmentation.** Fang X et al. In: *SPIE Medical Imaging 2020: Image Processing (SPIE MI 2020)* (Oral presentation)

Journal

- **Patient-specific reference model estimation for orthognathic surgical planning.** Fang X et al. In: *International journal of computer assisted radiology and surgery (IJCARs 2024)*(IPCAI Special Issue)
- **Large Language Models Diagnose Facial Deformity.** Lee J, Xu X, Kim D, Deng H.H, Kuang T, Lampen N, Fang X et al. In: *International journal of computer assisted radiology and surgery (IJCARs 2024)*(Under review)
- **Learning Soft Tissue Deformation from Incremental Simulations** Lampen N, Kim D, Xu X, Fang X et al. In: *Medical Physics (Medical Physics 2024)*.
- **Correspondence Attention for Facial Appearance Simulation.** Fang X et al. In: *Medical Image Analysis (MedIA 2024)*.
- **Shape Description Losses for Medical Image Segmentation.** Fang X et al. In: *Machine Vision and Applications (MVAP 2023)*
- **Deep learning for biomechanical modeling of facial tissue deformation in orthognathic surgical planning.** Lampen N, Kim D, Fang X et al. In: *International journal of computer assisted radiology and surgery (IJCARs 2022)*
- **Association of AI quantified COVID-19 chest CT and patient outcome.** Fang X et al. In: *International journal of computer assisted radiology and surgery (IJCARs 2021)*
- **Integrative analysis for COVID-19 patient outcome prediction.** Chao H*, Fang X*, Zhang J* et al. (*equal contribution) In: *Medical Image Analysis (MedIA 2021)*
- **Multi-organ segmentation over partially labeled datasets with multi-scale feature abstraction.** Fang X, and Yan P. In: *IEEE Transactions on Medical Imaging (IEEE TMI 2020)*.
- **Deep learning-based liver segmentation for fusion-guided intervention.** Fang X et al. In: *International journal of computer assisted radiology and surgery (IJCARs 2020)*
- **A method of rapid quantification of patient-specific organ doses for CT using deep-learning-based multi-organ segmentation and GPU-accelerated Monte Carlo dose computing.** Peng Z, Fang X et al. In: *Medical Physics (Medical Physics 2020)*

Abstract

- **Large Language Models Diagnose Facial Deformity.** Lee J, Xu X, Kim D, Deng H.H, Kuang T, Lampen N, Fang X et al. In: *Computer Assisted Radiology and Surgery (CARS 2024)* (Oral presentation)
- **Deep learning-based soft-tissue-driven craniomaxillofacial surgical planning.** Fang X et al. In: *Computer Assisted Radiology and Surgery (CARS 2023)* (Oral presentation)
- **Incremental Biomechanical Deep Learning Modeling of Facial Tissue Deformation.** Lampen N, Kim D, Xu X, Fang X et al. In: *Computer Assisted Radiology and Surgery (CARS 2023)* (Oral presentation)
- **Mandible landmark detection for craniomaxillofacial surgical planning using graph convolutional network.** Xu X, Kuang T, Chan A, Lampen N, Fang X et al. In: *Computer Assisted Radiology and Surgery (CARS 2023)* (Oral presentation)
- **A feasibility study on estimating desired postoperative face using deep learning for patients with craniomaxillofacial deformities** Lee J, Kim D, Xu X, Fang X et al. In: *Computer Assisted Radiology and Surgery (CARS 2023)* (Oral presentation)
- **Predicting facial changes following bony movement via deep correlation for orthogonal surgical planning.** Fang X et al. In: *Computer Assisted Radiology and Surgery (CARS 2022)* (Oral presentation)

RESEARCH PROJECTS

Unsupervised reference CMF model prediction for orthognathic surgical planning 07/2023 - 10/2023

- Proposed a deep query network (DQN) that employs similarity scores to aggregate the normal subjects in a high-dimensional space.
- Developed a self-supervised approach for reference CMF model prediction for patients with jaw deformities.

Foundation AI model for medical image analysis 04/2023 - 06/2023

- Developed an efficient 3D transformer as a backbone for medical image segmentation, surpassing the 3D ViT and Swin Transformer in both efficiency and accuracy.
- Worked on a self-supervised training approach for 3D transformers, to establish a foundational model in the medical imaging domain that can be applied to various tasks such as medical image classification, reconstruction, registration, and segmentation.

Soft-tissue driven orthognathic surgical planning 06/2022 - 03/2023

- Developed the first soft-tissue-driven approach for CMF surgical planning, substantially reducing the planning time by removing the need for repetitive guessing bony movement.
- Developed facial simulation module to qualitatively assess the effect of surgical plans on facial appearance, for virtual validation.

Facial simulation following bony movement for orthognathic surgical planning 05/2021 - 02/2022

- Developed an Attentive Correspondence assisted Movement Transformation network (ACMT-Net) with a novel CPSA module to estimate the change of one point set driven by the movement of another point set. The network leverages the local movement vector information by explicitly establishing the spatial correspondence between two point sets.
- The network achieved comparable accuracy to the state-of-the-art FEM simulation method, while substantially reducing computational time during the surgical planning.

Shape description losses for medical image segmentation 01/2021 - 04/2021

- Proposed several shape-based loss functions, e.g., bounding box, that facilitate the training of segmentation model by utilizing the crucial global geometrical and statistical information
- Investigated the effectiveness of shape descriptions with physical meanings in improving learning for segmenting different organs from various modalities.

COVID-19 patient outcome prediction 05/2020 - 12/2020

- Provided AI assisted severity scoring method, which was significantly more accurate than the current severity scoring by radiologists.
- Very first study that used holistic information of a patient including both imaging and non-imaging data for outcome prediction.

Multi-organ segmentation over partially labeled datasets 06/2019 - 04/2020

- Proposed a new pyramid-input and pyramid-output feature abstraction network (PIPO-FAN) that condense multi-scale features to reduce the semantic gaps between features from different scales.
- Proposed a target adaptive loss, which was integrated with a unified training strategy to enable image segmentation over multiple partially labeled datasets with a single model.

Liver segmentation in CT images using Deep CNN for fusion guided intervention 09/2018 - 05/2019

- Presented a deep learning based liver segmentation method for a new workflow of fusion guided intervention.
- The application of our developed registration technique was demonstrated through the fusion of various modalities with interventional computed tomography (CT).

SKILLS

Toolkits	PyTorch/TensorFlow/MXNet/Pycaffe, MLflow, Weights & Biases (W&B), SimpleITK, VTK, OpenCV, Matplotlib, Numpy, Pandas, Scipy, Scikit-learn, Open3d, PyTorch3D, 3D-Slicer, MIPAV, ParaView, ITK-SNAP, MeshLab, Pyinstrument, Protocol Buffers, Git, Version Control, Design Control, JIRA, Confluence, Claude Code
Programming	Python, C/C++, Matlab, R
Theories	Transformer, AI Agent, Foundation Model, Large Language Model, Diffusion model, Segment Anything Model, Point Cloud Analysis, Neural Radiance Field, Gaussian Splatting, Implicit Neural Networks, Domain Adaptation, Generative Adversarial Network, Graph Neural Network, Uncertainty Analysis, Self-supervised Learning, Multi-view Geometry
Applications	Instance Segmentation, Image Registration, Point Cloud Reconstruction/Registration, 360° Image Analysis, Image Captioning, Facial Simulation, Image Guided Surgery
Workflow	Requirements management, technical specifications, software design documents (SDD), experiment tracking, model registry/versioning, unit testing, verification & validation (V&V), cross-functional design reviews, MCP (Model Context Protocol) tool integration

AWARDS & HONORS

2024	The NVIDIA Bench-to-Bedside Award Runner-up
2022	Best of MICCAI in Computer Vision News
2022	MICCAI Society Young Scientist Award
2022	MICCAI Student Travel Award
2017	Ranked the top 12th among 94 students in Duke Tsinghua MLSS, with strong recommendation letter from Dr. Lawrence Carin, Vice Provost for Research, Duke University
2017	Meritorious Winner Mathematical Contest In Modeling (MCM)